Minggu Ke-3

Agile Methodology

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Fokus Pembelajaran

Mata kuliah Rekayasa Perangkat lunak (RPL), difokuskan pada Agile Metodology dalam proyek pengembangan perangkat lunak. Mata kuliah Rekayasa Perangkat Lunak membahas prinsip, konsep, metode, dan praktik dalam pengembangan perangkat lunak secara sistematis, terstruktur, dan terukur. Fokus pembelajaran diarahkan pada penerapan Agile Methodology, khususnya Scrum, dalam pengelolaan proyek sistem informasi berbasis tim.

Mata kuliah RPL tidak hanya mengajarkan cara membuat sistem, tetapi mengajarkan cara mengelola dan mengembangkan sistem secara profesional menggunakan Agile. Mahasiswa dilatih berpikir seperti:

1. System Analyst
2. IT Business Analyst
3. Software Engineer
4. Project Manager

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Fokus Pembelajaran

Kompetensi yang Dikembangkan

Hard Skills

1. Analisis kebutuhan sistem
2. Perancangan sistem
3. Manajemen proyek perangkat lunak

Soft Skills

1. Teamwork
2. Leadership
3. Problem solving
4. Adaptability
5. Communication

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Fokus Pembelajaran

Integrasi dalam Proyek Sistem Informasi 1

Mahasiswa diwajibkan:

1. Menggunakan pendekatan Agile-Scrum
2. Membagi proyek minimal 3 sprint
3. Melakukan presentasi hasil tiap sprint
4. Membuat dokumentasi ringan namun terstruktur

Artefak yang wajib dikumpulkan:

1. Vision Project
2. Product Backlog
3. Sprint Backlog
4. Burndown Chart
5. Increment System
6. Laporan Retrospective

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Agile Methodology

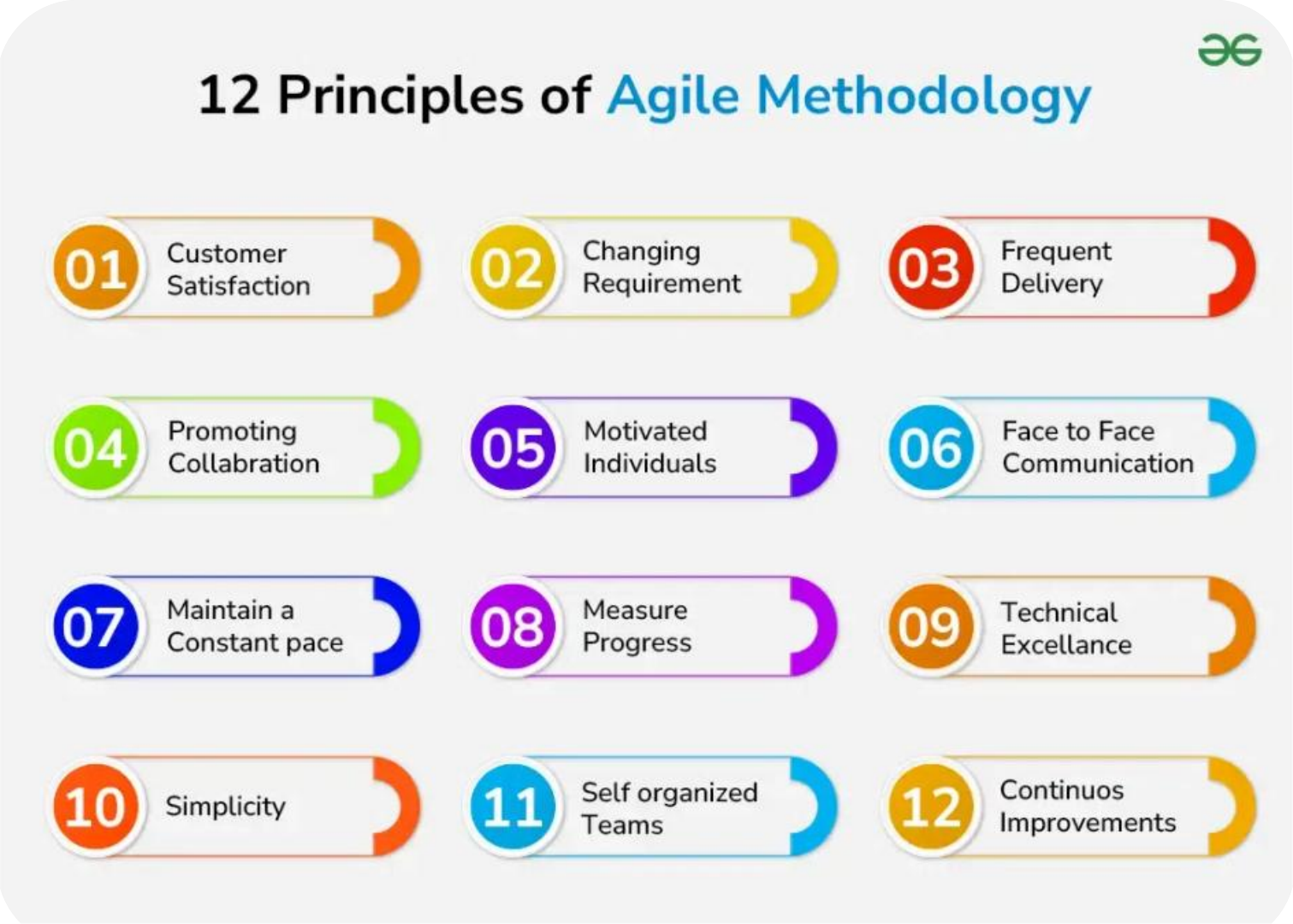
Agile Methodology is a way to manage projects by breaking them into smaller parts. It focuses on working together and making constant improvements. Teams plan, work on the project, and then review how things are going in a repeating cycle.

1. It focuses on delivering smaller pieces of work regularly instead of one big launch.
2. This allows teams to adapt to changes quickly and provide customer value faster.
3. Major companies like Facebook, Google, and Amazon use Agile because of its adaptability and customer-focused approach.
4. Prioritize flexibility, collaboration, and customer satisfaction.

Sumber Referensi : Klik untuk akses Agile Methodology - GeeksforGeeks

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Principles of Agile Methodology



12 Principles for Agile Methodology

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Principles of Agile Methodology

The Agile Alliance defines twelve lightness principles for those who need to attain agility:

1. Our highest priority is to satisfy the client through early and continuous delivery of valuable computer software.
2. Welcome dynamic necessities, even late in development. Agile processes harness modification for the customer's competitive advantage.
3. Deliver operating computer software often, from a pair of weeks to a couple of months, with a preference to the shorter timescale.
4. Business individuals and developers should work along daily throughout the project.
5. The build comes around actuated people. offer them the setting and support they have, and trust them to urge the task done.
6. the foremost economical and effective methodology of conveyancing info to and among a development team is face-to-face speech.

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Principles of Agile Methodology

The Agile Alliance defines twelve lightness principles for those who need to attain agility:

7. Working with computer software is the primary life of progress.
8. Agile processes promote property development. The sponsors, developers, and users will be able to maintain a relentless pace indefinitely.
9. Continuous attention to technical excellence and smart style enhances nimbleness.
10. Simplicity—the art of maximizing the number of work not done—is essential.
11. the most effective architectures, necessities, and styles emerge from self–organizing groups.
12. At regular intervals, the team reflects on a way to become simpler, then tunes and adjusts its behavior consequently.

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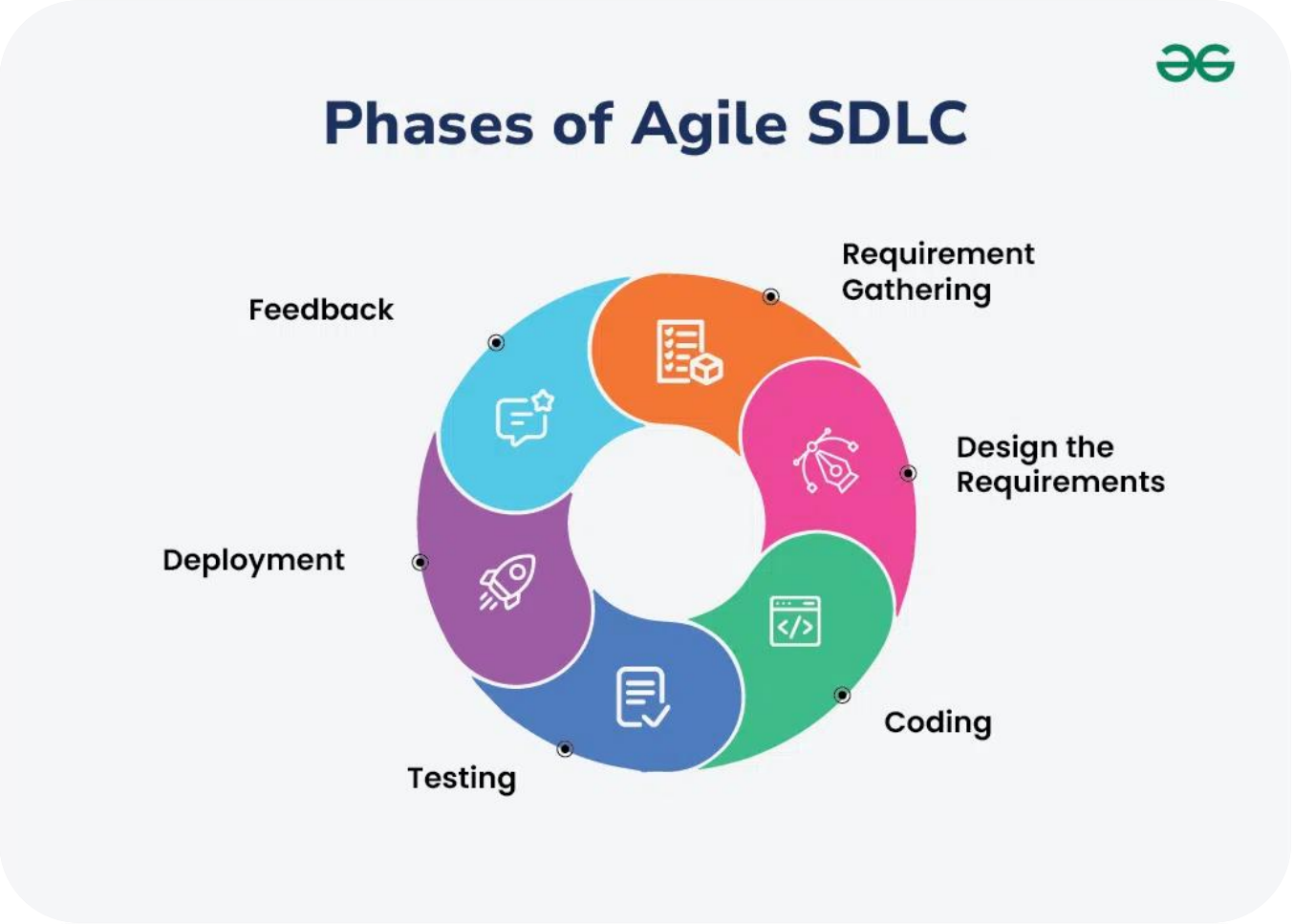
Agile SDLC (Software Development Life Cycle)

Software Development Life Cycle (SDLC) is a process of maintaining or building software applications/services/systems. Generally, it includes various levels, from initial development plan and analysis to post-development software testing and evaluation. It also consists of the models and methodologies that development teams use to develop the software systems. An information system is designed to perform a particular set of assignments. Frequently, this set of tasks that the system will perform offers well-defined results, which include complex computation and processing.

It is a hard and boring job to govern the entire development process to ensure that the end product comprises a high degree of integrity and strength, as well as user acceptance. Thus, a systematic development process that can point to the understanding of the scope and complication of the total development process is essential to achieve the characteristics of a successful system.

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Agile SDLC (Software Development Life Cycle)



Sumber: klik untuk akses Agile SDLC (Software Development Life Cycle) - GeeksforGeeks

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Steps of Agile SDLC Model

1. **Requirement Gathering**. In requirement Gathering collaboration takes place with stakeholders to understand and prioritize needs for the project, that focus on delivering value. In this step techniques like user stories and workshops for effective communication.

2. **Design the Requirements**. In design the requirement step that gathered requirements is translated into actionable tasks, breaking them down into smaller, manageable chunks. In this step developers develop visual representations of the solution, such as wireframes or prototypes, to get early feedback and ensure alignment with stakeholder expectations.

3. **Coding.** During the coding step, the development team implements the software in short, iterative cycles known as sprints. Each sprint focuses on delivering small, functional increments of the product. Developers write code, continuously integrate it into the main codebase, and regularly commit changes to detect and address issues early.

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Steps of Agile SDLC Model

4. **Testing / Quality Assurance**. Testing is an integral part of each iteration, ensuring the quality and functionality of the software. Unit tests are written to verify individual components, while integration tests check that different parts of the system work together as intended. User Acceptance Testing (UAT) involves end-users testing the software in real-world scenarios to ensure it meets their needs.

5. **Deployment**. Deployment involves releasing the software increments to production frequently and reliably. Automated deployment tools makes the process, allowing for quick and consistent releases. Monitoring the deployment process and system performance helps identify and resolve any issues promptly.

6. **Feedback**. Feedback is crucial for continuous improvement in Agile. Stakeholders and end-users provide feedback through surveys, direct communication, and usage analytics. This feedback is used to refine requirements, prioritize changes, and identify areas for improvement.

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Agile SDLC Models

Six methods are identified as agile development methods they are following :

- Crystal methodologies
- Dynamic software development method
- Feature-driven development
- Lean software development
- Scrum
- Extreme programming

Pembahasan detail masing-masing model dapat dilakukan dengan cara klik pada link diatas

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Popular Agile Tools for Software Development

Agile Tool for software development or platforms that help teams manage and track their Agile projects more effectively. These tools streamline collaboration, enhance productivity, and ensure that projects stay on track.

- Jira
- Trello
- Asana
- Pivotal Labs
- ClickUp
- Monday.com
- CollabNet
- Axosoft
- Zoho Sprints
- Active Collab

1. Pembahasan pada link diatas cukup Jira dan Trello
2. Belajar mandiri, secara individu mahasiswa mengenal Trello melalui berbagai media seperti:

https://www.cloudwards.net/trello-beginners-guide/

Tutorial Trello : https://www.youtube.com/watch?v=Jx9a9gb7-Uw

Apa itu Trelo? https://www.domainesia.com/tips/trello/#Apa_itu_Trello

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Case study

Dikerjakan dengan kelompok masing-masing dengan metode pembalajaran: Case Study, Collaborative Learning:

1. Mahasiswa telah mempelajari beberapa referensi pada slide sebelumnya
2. Praktikan case study berikut di minggu ke 3 dan akan didemokan pada minggu ke 4
3. Link handout case study dapat diunduh : Proyek Sistem Operasi Matematika

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